Academic Integrity and AI-Tool Usage Declaration

Project: AUTOSAR HLD Document Analysis Assistant (Case Study 1)

Student: ANBU SELVAN M

Register number: 127003017

University: SASTRA Deemed University

Declaration 1: Academic integrity

I declare that:

1. This submission is my original work for this project. I used the AI tools listed in Declaration 2, and I am accountable for the correctness of everything I submit and can explain all of it.
2. All external code, models and documents I used are cited in section 9 of the technical report. The test document is synthetic and was generated for this project. No confidential or third-party HLD was used.
3. No result is fabricated. Every figure in the report comes from a run of the code in this submission, and the result files are in the Evaluation_Results folder.
4. My individual contribution is stated accurately: this is an individual project, and the work described in the report is mine, with the AI assistance declared below.

Signature:  Date: 13-10-2026

Declaration 2: AI-tool usage

Each AI tool or pre-trained model used, including for coding assistance, is listed with its purpose, the artifact it affected, and how I verified the output.

AI Model/Tool	Purpose	Artifact affected	Verified by student
Claude (Anthropic)	Planned the workflow. Drafted the notebook code, the synthetic HLD generator, the routing and citation-check code, the API, web app and Docker files, the test questions, and the report, synopsis, user guide and declaration text. Proposed grades for the question-answering and flow-summary evaluations	Code folder (notebook, qa_router.py, hld_core.py, api.py, app.py, generate_synthetic_hld.py, Dockerfile, docker-compose.yml, README), Input_Data (synthetic HLD, questions, ground truth), Evaluation_Results grades, Documentation, Synopsis, Declarations	Ran every notebook cell and checked its output. Tested the web app and the Docker containers. Checked graded answers against the HLD PDF. Read the report
Qwen2.5 3B (served by Ollama)	Extracting components and ports from the HLD. Answering open questions. Writing the flow summaries that were evaluated. Splitting and judging answers in the faithfulness sample	SQLite component and port rows. Answers in the question-answering result files. Flow summaries. Faithfulness results	Compared extracted rows with the labelled ground truth (precision and recall 1.0). Graded answers and summaries by hand. Read the faithfulness verdicts
BAAI/bge-small-en-v1.5	Embedding document chunks and questions for semantic search	ChromaDB index	Compared retrieval methods on 10 questions (recall at 3). Checked top results by hand
cross-encoder/ms-marco-MiniLM-L-6-v2	Reranking retrieved chunks and scoring relevance for the refusal cutoff	Retrieval results and refusal decisions	Same recall comparison. Tested out-of-scope questions and recorded one wrong refusal
Tesseract OCR	Fallback text recognition for scanned	Ingestion step	Not exercised on the test documents, which

AI Model/Tool	Purpose	Artifact affected	Verified by student
	pages		contain no scanned pages

No other AI tool or pre-trained model was used. I remain accountable for the correctness of all submitted work and can explain it.

Signature: 

Date: 13-10-2026